

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I B. Charthanya Prakash Reddy (1924255) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Charthanya
Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Charthanya

Q1 Topic: IPv4 subnetting (192.168.100.0/24)

- School of Computing (125)
 - Network Address: 192.168.100.0
 - Broadcast Address: 192.168.100.127
 - Host Range: 192.168.100.1 - 192.168.100.126
 - Usable Host: 126.
- School of Electronics (126)
 - Network Address: 192.168.100.128
 - Broadcast Address: 192.168.100.191
 - Host Range: 192.168.100.129 - 192.168.100.190.
 - Usable Hosts: 62.
- School of Mechanical Engineering (127).
 - Network Address: 192.168.100.192.
 - Broadcast Address: 192.168.100.223.
 - Host Range: 192.168.100.193
 - Usable Hosts: 30.

• Remaining unused IP Range:

• 192.168.100.224 - 192.168.100.255.

Q2.

Topic: IPv4 Subnetting (10.10.0.0/16).

- Software Development (124)
 - Network Address: 10.10.0.0

- Broadcast Address : 10.10.0.255
- Host Range : 10.10.0.1 - 10.10.0.254
- Usable Hosts : 254.
- Quality Assurance (125)
 - Network Address : 10.10.1.128
 - Broadcast : 10.10.1.191
 - Host Range : 10.10.1.129 - 10.10.1.190
 - Usable Hosts : 62.
- Executive Management (127)
 - Network Address : 10.10.1.192
 - Broadcast Address : 10.10.1.223.
 - Host Range : 10.10.1.193 - 10.10.1.222
 - Usable Hosts : 30.
- Remaining unused IP Range :
 - 10.10.1.224 - 10.10.255.255.

Q3:

Topic : IPv6

Given IPv6 Address:

1. • 2001:db8:abcd::/64

2. Expanded : 2001:0db8:abcd:0000:0000:0000:0000/64

3. Network Prefix:

• 2001:0bd8:abcd::/64

4. Number of Host Addresses:

• $2^{64} = 18,446,744,073,709,551,616$

5. Total Possible Host Addresses:

• 18,446,744,073,709,551,616 (2^{64})

Q 4.
Topic: Routing.

Routing table: • 192.168.1.0/24

• ~~192.168.2.0/24~~

• 192.168.3.0/24

Destination IP: 192.168.2.45

1. Destination Subnet:

• 192.168.2.0/24

2. Matching Routing entry:

• 192.168.2.0/24

3. Next - Hop network

• The router forwards the packet through the interface connected to the 192.168.2.0/24 network.

4. Default Route:

• 0.0.0.0/0.

Q5.

Topic: LAN Configuration.

Administration LAN (/26)

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63.
- Host Range: 192.168.10.1 - 192.168.10.62.
- Three Computer IPs:
 - 192.168.10.2
 - 192.168.10.3
 - 192.168.10.4
- Default Gateway:
 - 192.168.10.1

Finance LAN (/26)

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127
- Host Range: 192.168.10.65 - 192.168.10.126
- These Computer IPs: 192.168.10.66
 - 192.168.10.67
 - 192.168.10.68.
- Default Gateway: 192.168.10.68. //